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Smart electricity Billing System
problem statement:

Ensure electricity usage is calculated and billed correctly.

Tasks:

1.develop billing calculation logic
2.write Junit tests for tariff calculations

3. implement CI/CD pipeline with automated testing

- Set up version control using github
- Configure CI pipeline (e.g Jenkins/Github actions) to trigger on code commits
- Automate Build Process using maven
- Integrate automated unit test in pipeline
- Configure CD pipeline for automatic deployment to Staging/production

4. Deploy service on docker and Kubernetes

- Containerize application using docker(write dockerfile)
- Build and push docker image to container registry (Docker hub)
- Create Kubernetes deployment and service YAML files

Java code

```
package com.demo;

public class ElectricityBilling {

  private static final double FIXED_CHARGE = 50.0;

  public double calculateBill(int units) {
    if (units < 0) {
      throw new IllegalArgumentException("Units consumed cannot be negative.");
    }
    double charge = 0.0;
    if (units <= 100) {
      charge = units * 1.50;
    } else if (units <= 200) {
      charge = (100 * 1.50) + ((units - 100) * 2.50);
    } else if (units <= 500) {
      charge = (100 * 1.50) + (100 * 2.50) + ((units - 200) * 4.00);
    } else {
      charge = (100 * 1.50) + (100 * 2.50) + (300 * 4.00) + ((units - 500) * 6.00);
    }
    return charge + FIXED_CHARGE;
  }
```

```
    public String getBillSummary(String customerName, int units) {
        if (customerName == null || customerName.trim().isEmpty()) {
            throw new IllegalArgumentException("Customer name cannot be empty.");
        }
        double total = calculateBill(units);
        return "Customer: " + customerName + " | Units: " + units + " | Bill: Rs." + total;
    }

    public static void main(String[] args) {
        System.out.println("=== Smart Electricity Billing System ===");
        ElectricityBilling billing = new ElectricityBilling();
        System.out.println(billing.getBillSummary("Alice", 80));
        System.out.println(billing.getBillSummary("Bob", 150));
        System.out.println(billing.getBillSummary("Charlie", 350));
        System.out.println(billing.getBillSummary("David", 600));
    }
}
```

Junit program testcases

```
package com.demo;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class ElectricityBillingTest {

    private ElectricityBilling billing;

    @BeforeEach
    void setUp() { billing = new ElectricityBilling(); }

    @Test
    void testSlab1Under100Units() {
        assertEquals(125.0, billing.calculateBill(50), 0.01);
    }

    @Test
    void testSlab1Exactly100Units() {
        assertEquals(200.0, billing.calculateBill(100), 0.01);
    }

    @Test
    void testSlab2Between101And200Units() {
```

```
    assertEquals(325.0, billing.calculateBill(150), 0.01);
}

@Test
void testSlab2Exactly200Units() {
    assertEquals(450.0, billing.calculateBill(200), 0.01);
}

@Test
void testSlab3Between201And500Units() {
    assertEquals(1050.0, billing.calculateBill(350), 0.01);
}

@Test
void testSlab4Above500Units() {
    assertEquals(2250.0, billing.calculateBill(600), 0.01);
}

@Test
void testZeroUnits() {
    assertEquals(50.0, billing.calculateBill(0), 0.01);
}

@Test
void testNegativeUnitsThrowsException() {
    assertThrows(IllegalArgumentException.class, () -> billing.calculateBill(-10));
```

```
    }

    @Test
    void testBillSummaryContainsCustomerName() {
        String summary = billing.getBillSummary("Alice", 50);
        assertTrue(summary.contains("Alice"));
        assertTrue(summary.contains("125.0"));
    }

    @Test
    void testEmptyCustomerNameThrowsException() {
        assertThrows(IllegalArgumentException.class, () -> billing.getBillSummary("", 100));
        assertThrows(IllegalArgumentException.class, () -> billing.getBillSummary(null, 100));
    }
}
```

Java program development and testing

Mvm clean test

```
Problems   Output   Debug Console   Terminal   Ports

PS C:\Users\sabari\Desktop\ElectricityBilling> mvn clean test
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.ElectricityBillingTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.130 s -- in com.demo.ElectricityBillingTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.075 s
[INFO] Finished at: 2026-04-15T18:22:28+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\sabari\Desktop\ElectricityBilling>
PS C:\Users\sabari\Desktop\ElectricityBilling>
```

Java program development and testing

Mvn clean package

```
Problems   Output   Debug Console   Terminal   Ports

PS C:\Users\sabari\Desktop\ElectricityBilling>  mvn clean package
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ electricitybilling ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sabari\Desktop\ElectricityBilling\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ electricitybilling ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ electricitybilling ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sabari\Desktop\ElectricityBilling\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ electricitybilling ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ electricitybilling ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.ElectricityBillingTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.152 s -- in com.demo.ElectricityBillingTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ electricitybilling ---
[INFO] Building jar: C:\Users\sabari\Desktop\ElectricityBilling\target\electricitybilling-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.646 s
[INFO] Finished at: 2026-04-15T18:24:16+05:30
[INFO] ------------------------------------------------------------------------
```

Java program development and testing

java -jar target/electricitybilling-1.0.jar

```
PS C:\Users\sabari\Desktop\ElectricityBilling> java -jar target/electricitybilling-1.0.jar
>>
=== Smart Electricity Billing System ===
Customer: Alice | Units: 80 | Bill: Rs.170.0
Customer: Bob | Units: 150 | Bill: Rs.325.0
Customer: Charlie | Units: 350 | Bill: Rs.1050.0
Customer: David | Units: 600 | Bill: Rs.2250.0
PS C:\Users\sabari\Desktop\ElectricityBilling>
ava: Ready
```

Github push

git init

git add .

git commit -m "Initial commit"

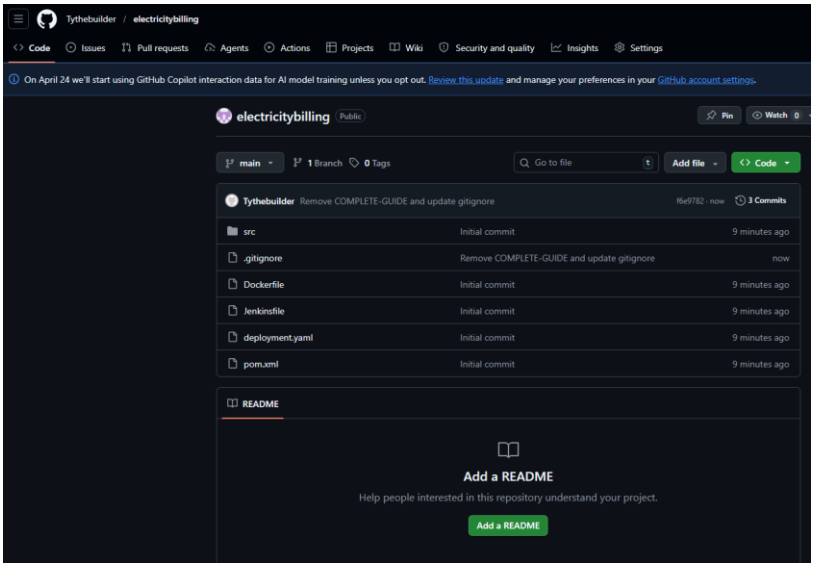
git branch -M main

git remote add origin https://github.com/Tythebuilder/ElectricityBilling.git

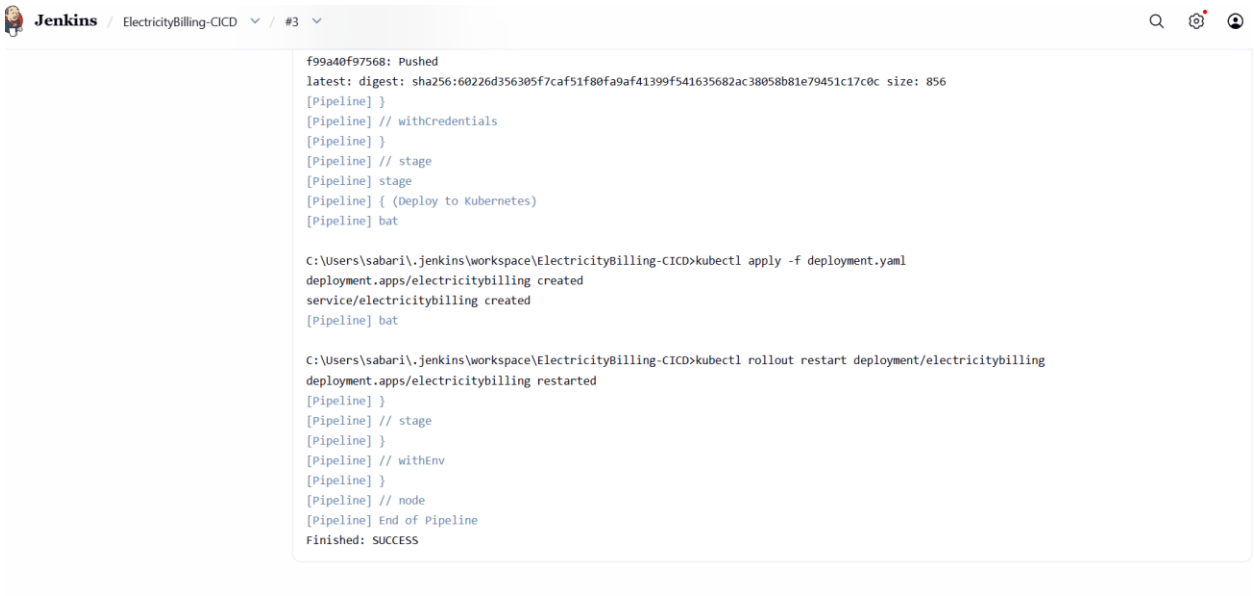
git push -u origin main

```
Problems   Output   Debug Console   Terminal   Ports                                   powershell

PS C:\Users\sabari\Desktop\ElectricityBilling> cd C:\Users\sabari\Desktop\ElectricityBilling
PS C:\Users\sabari\Desktop\ElectricityBilling> git init
Reinitialized existing Git repository in C:/Users/sabari/Desktop/ElectricityBilling/.git/
PS C:\Users\sabari\Desktop\ElectricityBilling> git add .
PS C:\Users\sabari\Desktop\ElectricityBilling> git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
PS C:\Users\sabari\Desktop\ElectricityBilling> git branch -M main
PS C:\Users\sabari\Desktop\ElectricityBilling> git remote add origin https://github.com/Tythebuilder/ElectricityBilling.git
PS C:\Users\sabari\Desktop\ElectricityBilling> git push -u origin main
Enumerating objects: 36, done.
Counting objects: 100% (36/36), done.
Delta compression using up to 8 threads
Compressing objects: 100% (21/21), done.
Writing objects: 100% (36/36), 7.86 KiB | 893.00 KiB/s, done.
Total 36 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
remote: This repository moved. Please use the new location:
remote:   https://github.com/Tythebuilder/electricitybilling.git
To https://github.com/Tythebuilder/ElectricityBilling.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\sabari\Desktop\ElectricityBilling>
```

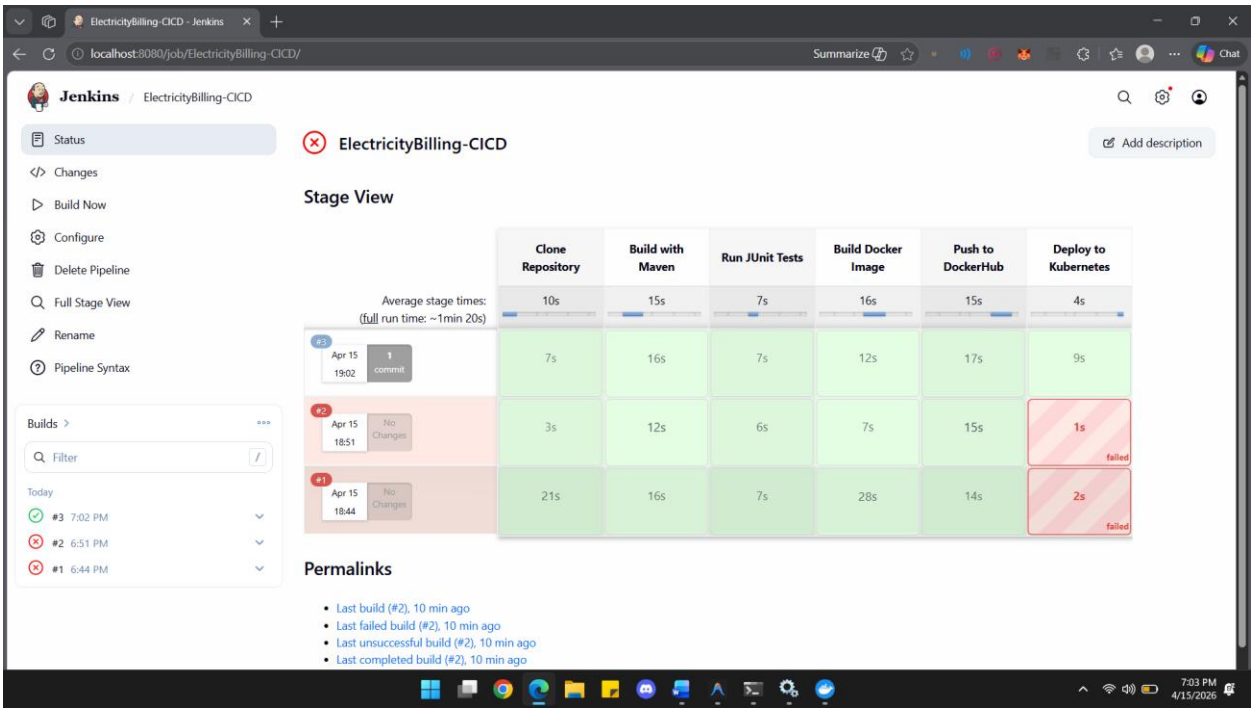
Github imageimage



Jenkin cicd pipeline creation and execution



Jenkin cicd pipeline creation and execution



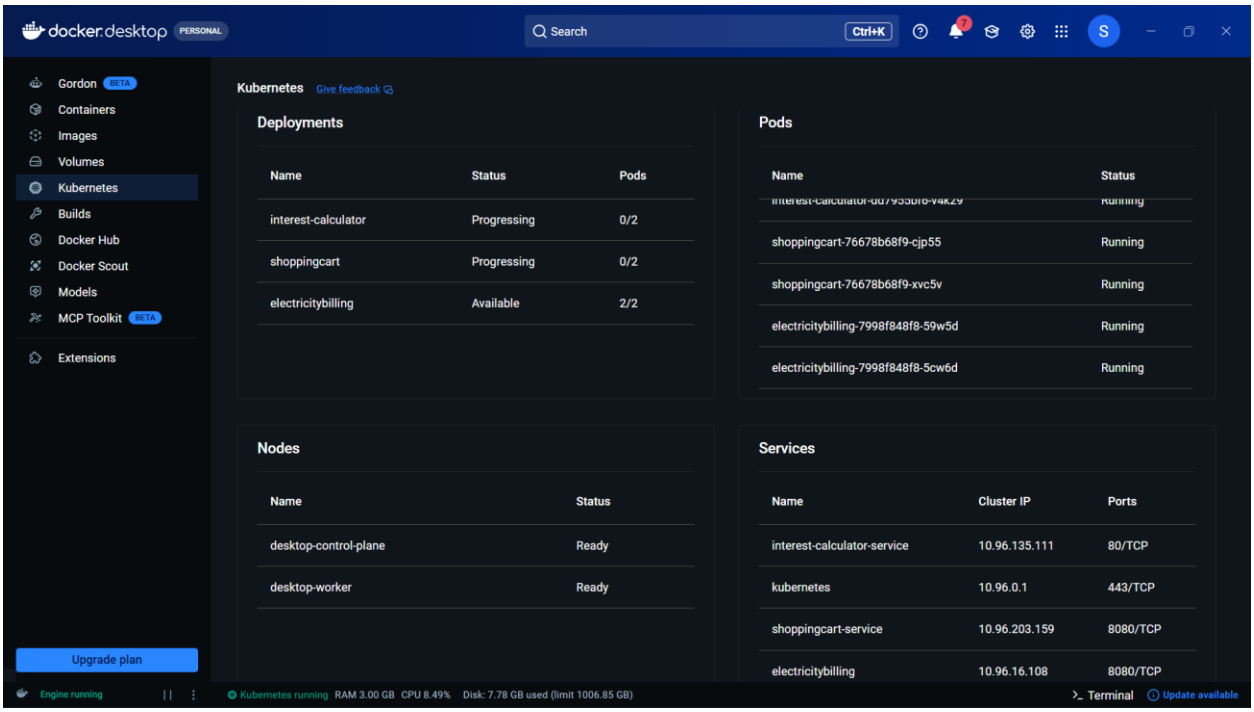
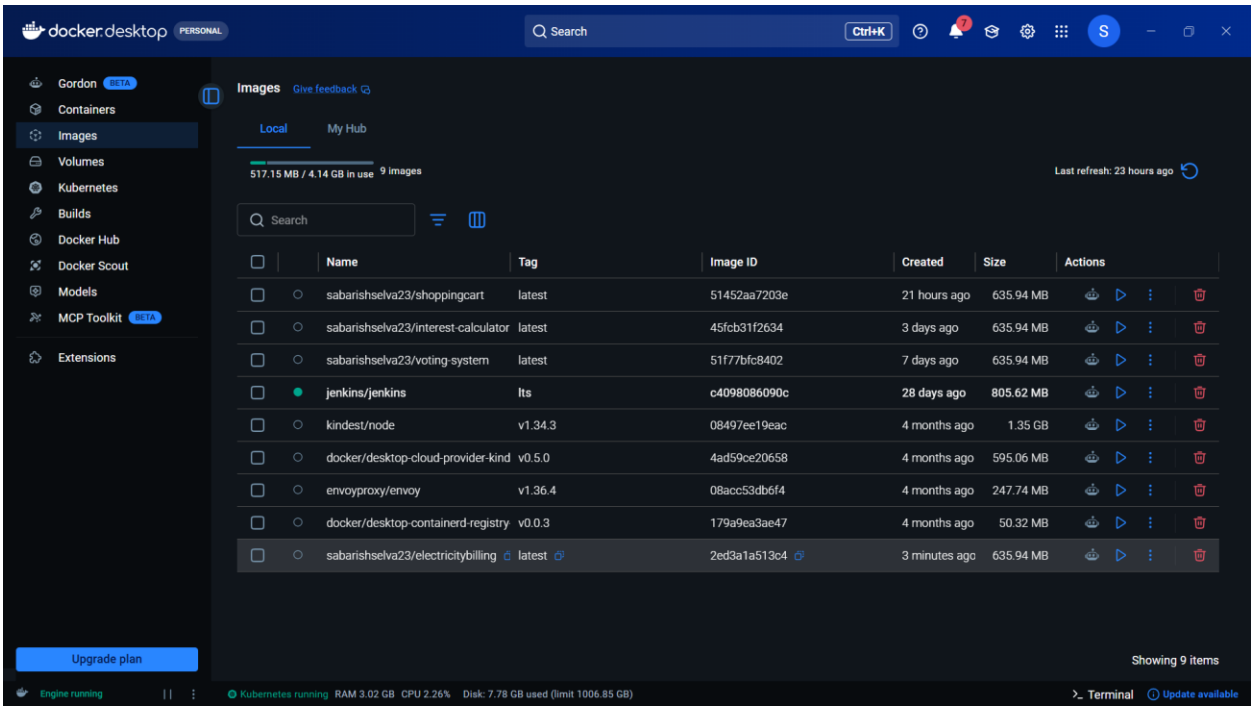
Kubernetes and docker service

```
Explorer
ElectricityBilling
  src
    main\java\com...
      ElectricityBilling....
    test\java\com\...
      ElectricityBilling...
  target
  .gitignore
  COMPLETE-GUIDE....
  deployment.yaml
  Dockerfile
  Jenkinsfile
  pom.xml
Outline
Timeline
Java Projects
Maven

Problems   Output   Debug Console   Terminal   Ports                                   powershell

                                                    mvn clean package
>> docker build -t sabarishselva23/electricitybilling:latest .
>>
[INFO] -------------------------------------------------------
[INFO] Running com.demo.ElectricityBillingTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.131 s -- in com.demo.ElectricityBillingTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ electricitybilling ---
[INFO] Building jar: C:\Users\sabari\Desktop\ElectricityBilling\target\electricitybilling-1.0.jar
[INFO] ------------
[INFO] BUILD SUCCESS   docker-desktop://dashboard/build/desktop-linux/desktop-
[INFO] ------------    linux/uk8cuwjhmn6ofpqhknhj2ovkq (ctrl + click)
[INFO] Total time:  7.765 s
[INFO] Finished at: 2026-04-15T19:16:04+05:30
[INFO] ------------------------------------------------------------------------
[+] Building 7.1s (8/8) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                0.1s
 => => transferring dockerfile: 326B                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                          1.6s
 => [internal] load .dockerignore                                                                   0.1s
 => => transferring context: 2B                                                                     0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430   0.1s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430   0.1s
 => [internal] load build context                                                                   0.3s
 => => transferring context: 3.35kB                                                                 0.3s
 => CACHED [2/3] WORKDIR /app                                                                       0.0s
 => [3/3] COPY target/electricitybilling-1.0.jar app.jar                                            2.0s
 => exporting to image                                                                              2.1s
 => => exporting layers                                                                             0.8s
 => => exporting manifest sha256:8aa17f9f99762c86eecf8b8c475f5f7cd24a19570b872a22a22e5f8cc80a1012   0.1s
 => => exporting config sha256:559474f558cf0697dd06f66a0eaf6183c651fd4d2d7a678e8605662896697568    0.1s
 => => exporting attestation manifest sha256:c7bcbd24b2576139e108cb8ca4b2faf136fef36be024e21fed29d86ecebd0c91   0.2s
 => => exporting manifest list sha256:2ed3a1a513c43ab5fa1357fc0cb0b841b1481f2ba41169e9b7a08832b091909b   0.3s
 => => naming to docker.io/sabarishselva23/electricitybilling:latest                                0.0s
 => => unpacking to docker.io/sabarishselva23/electricitybilling:latest                             0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/lwtrrkx9fstd412431lvt77l4
PS C:\Users\sabari\Desktop\ElectricityBilling>

main   ⊗ 0 ⚠ 0   Java: Ready   Tythebuilder (20 hours ago)   Ln 31, Col 23   Spaces: 2   UTF-8   CRLF   {} YAML   Line 167   Count 5414   Antigravity - Se
```

Kubernetes and docker service

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/lwtrrkx9fstd412431lvt77l4
PS C:\Users\sabari\Desktop\ElectricityBilling> kubectl apply -f deployment.yaml
deployment.apps/electricitybilling unchanged
service/electricitybilling unchanged
PS C:\Users\sabari\Desktop\ElectricityBilling> kubectl get pods
NAME                                  READY   STATUS             RESTARTS        AGE
electricitybilling-7998f848f8-59w5d   1/1     Running            0               15m
electricitybilling-7998f848f8-5cw6d   1/1     Running            0               14m
interest-calculator-dd7955bf6-d2lbc   0/1     CrashLoopBackOff   77 (3m2s ago)   2d19h
interest-calculator-dd7955bf6-v4k29   0/1     CrashLoopBackOff   77 (2m40s ago)  2d19h
shoppingcart-76678b68f9-cjp55         0/1     CrashLoopBackOff   36 (2m52s ago)  21h
shoppingcart-76678b68f9-xvc5v         0/1     CrashLoopBackOff   36 (2m53s ago)  21h
PS C:\Users\sabari\Desktop\ElectricityBilling>

va: Ready                                   Tythebuilder (20 hours ago)   Ln 31, Col 23   Spaces: 2
```

Kubernetes and docker service



Kubernetes and docker service